

CS310 Natural Language Processing

Assignment 2 (written)

Total points: 10

Consider the learning objective of word2vec with SkipGram architecture: to predict a probability of an outside word o given a specific center word c , and the probability is modeled by a naïve softmax function:

$$P(o|c) = \frac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)}$$

Here $\mathbf{u}_o, \mathbf{v}_c \in \mathbb{R}^d$ are the word vectors/embeddings to be learned, and "Vocab" indicates the entire vocabulary. The parameters for all words in the vocabulary are stored in two matrices, $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d \times |\text{Vocab}|}$. For example, the o -th column in $\mathbf{U}$ is the outside vector $\mathbf{u}_o$, and the c -th column in $\mathbf{V}$ is the center vector $\mathbf{v}_c$.

1)

Suppose for a particular center word c and outside word o , let $\mathbf{y}$ be the true distribution and $\hat{\mathbf{y}}$ be the predicted distribution. Both $\mathbf{y}$ and $\hat{\mathbf{y}}$ are vectors with length equal to the vocabulary size. For $\mathbf{y}$, the o -th entry is $y_o = 1$ and all other entries $y_w = 0$ ($w \neq o$); for $\hat{\mathbf{y}}$, the o -th entry $\hat{y}_o$ is a scalar representing the probability $P(o|c)$, and $\hat{y}_w$ at w -th entries ($w \neq o$) are probabilities for other outside words.

Show that the cross-entropy loss $J(\mathbf{y}, \hat{\mathbf{y}}) = -\log(\hat{y}_o)$. You may describe your answer in words.

Answer

The cross-entropy loss is $J(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{w \in \text{Vocab}} y_w \log(\hat{y}_w)$.

Since $\mathbf{y}$ is a one-hot vec, $y_o = 1$ and $y_w = 0$ for all $w \neq o$,

simplifies where $w = o$:

$$J(\mathbf{y}, \hat{\mathbf{y}}) = -(1 \cdot \log(\hat{y}_o) + \sum_{w \neq o} 0 \cdot \log(\hat{y}_w)) = -\log(\hat{y}_o)$$

(1 points)

2)

The cross-entropy loss J is a function of $\mathbf{v}_c$, o , and $\mathbf{U}$, because $\hat{\mathbf{y}}$ are computed from them, and $\mathbf{y}$ is determined by o .

Compute the partial derivative $\frac{\partial J}{\partial \mathbf{v}_c}$. Please write the final answer in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{U}$. Do not use specific elements like $\mathbf{u}_o$ or $\hat{y}_o$.

...exp(u_o[⊤]v_c) + exp(u_w[⊤]v_c).

Answer

expand the loss funct:

$$J = -\log \left(\frac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_w \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \right) = -\mathbf{u}_o^\top \mathbf{v}_c + \log \sum_w \exp(\mathbf{u}_w^\top \mathbf{v}_c)$$

partial derivative w.r.t. $\mathbf{v}_c$:

$$\frac{\partial J}{\partial \mathbf{v}_c} = -\mathbf{u}_o + \sum_w \frac{\exp(\mathbf{u}_w^\top \mathbf{v}_c)}{\sum_x \exp(\mathbf{u}_x^\top \mathbf{v}_c)} \mathbf{u}_w = -\mathbf{u}_o + \sum_w \hat{y}_w \mathbf{u}_w$$

$\mathbf{u}_o = \mathbf{U}\mathbf{y}$ and $\sum_w \hat{y}_w \mathbf{u}_w = \mathbf{U}\hat{\mathbf{y}}$. Thus:

$$\frac{\partial J}{\partial \mathbf{v}_c} = \mathbf{U}(\hat{\mathbf{y}} - \mathbf{y})$$

(5 points)

3)

Compute the partial derivatives of J with respect to each of the outside word vectors $\mathbf{u}_w$. There will be two cases: $w = o$ for the true outside word; $w \neq o$ for other words. Please write the final answer in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{v}_c$, and you can use subscript (e.g., y_w).

Answer

Using the expanded form of J , consider two cases for $\mathbf{u}_w$:

1. For $w = o$:

$$\frac{\partial J}{\partial \mathbf{u}_o} = -\mathbf{v}_c + \frac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_x \exp(\mathbf{u}_x^\top \mathbf{v}_c)} \mathbf{v}_c = (\hat{y}_o - 1)\mathbf{v}_c$$

2. For other words $w \neq o$:

$$\frac{\partial J}{\partial \mathbf{u}_w} = 0 + \frac{\exp(\mathbf{u}_w^\top \mathbf{v}_c)}{\sum_x \exp(\mathbf{u}_x^\top \mathbf{v}_c)} \mathbf{v}_c = \hat{y}_w \mathbf{v}_c$$

Since $y_o = 1$ and $y_w = 0$ for $w \neq o$, both cases can be unified as:

$$\frac{\partial J}{\partial \mathbf{u}_w} = (\hat{y}_w - y_w)\mathbf{v}_c$$

(4 points)